

TF-1: Fine-Tuning a Transformer for Text Classification

Transformer family · CORE · 25 points

1. Problem & Data

This project fine-tunes a pretrained Transformer encoder for text classification and compares it against a classical baseline. The dataset is AG News, a four-class topic classification benchmark (World, Sports, Business, Sci/Tech) with 120,000 training examples and 7,600 test examples, obtained from the Hugging Face Hub.

2. Method

Two model families were trained and evaluated on the same fixed AG News test set (7,600 examples):

- TF-IDF + Logistic Regression — a classical baseline (max 20,000 features, unigrams and bigrams, scikit-learn LogisticRegression).
- DistilBERT (distilbert-base-uncased) — fine-tuned end-to-end via the Hugging Face Trainer API, learning rate $2e-5$, batch size 16, weight decay 0.01.

Both models were trained from scratch on each training subset described below, using the same fixed random seed (42) for data splitting and initialization, so that the comparison at each data size is apples-to-apples.

3. Original Contribution: Data-Efficiency Curve

Hypothesis: A pretrained Transformer needs substantially less labeled data than a classical bag-of-words baseline to reach comparable accuracy, because its representations already encode general language structure from pretraining. If true, the accuracy gap between DistilBERT and TF-IDF should be largest at small training-set sizes and shrink as more labeled data becomes available.

To test this, both models were trained on six increasing fractions of the AG News training set — 1%, 5%, 10%, 25%, 50%, and 100% (1,200 to 120,000 examples) — and evaluated on the same held-out test set at every fraction. To keep the comparison fair, DistilBERT was given proportionally more training epochs at the smallest fractions (few-shot fine-tuning typically requires more passes over a small set to converge), while TF-IDF, being a single-pass convex fit, needed no such adjustment.

4. Results

4.1 Data-Efficiency Curve

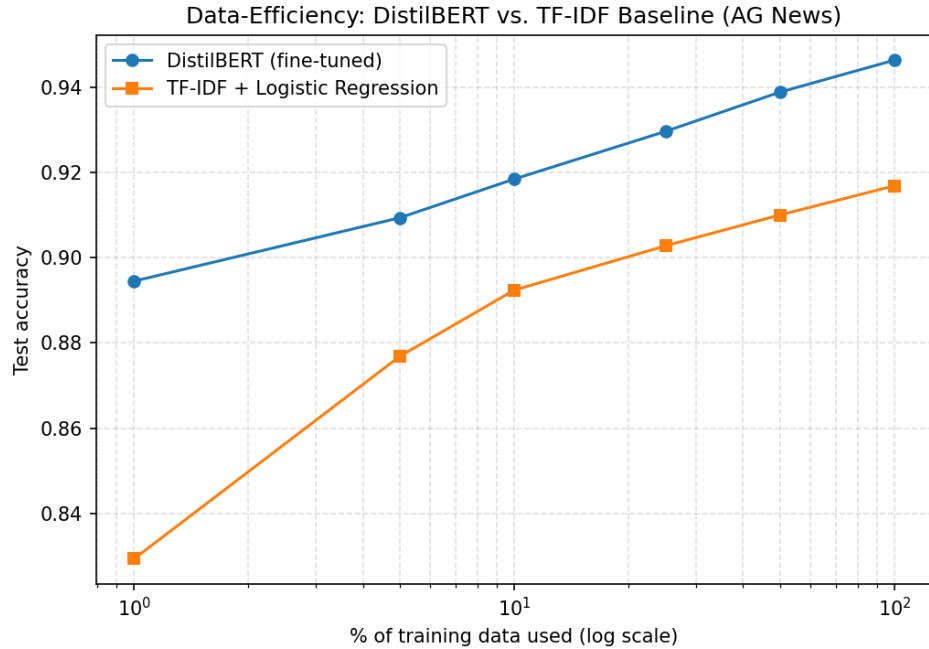


Figure 1. Test accuracy vs. percentage of training data used (log scale), comparing fine-tuned DistilBERT against a TF-IDF + Logistic Regression baseline on AG News.

4.2 Numerical Results

Fraction	Train Examples	TF-IDF + LogReg Acc.	DistilBERT Acc.	Gap (pp)
1%	1,200	82.93%	89.45%	+6.52
5%	6,000	87.68%	90.93%	+3.25
10%	12,000	89.24%	91.84%	+2.60
25%	30,000	90.28%	92.96%	+2.68
50%	60,000	91.00%	93.88%	+2.88
100%	120,000	91.68%	94.63%	+2.95

DistilBERT outperformed the TF-IDF baseline at every data fraction. Critically, the size of that advantage was not constant: the gap was 6.52 percentage points at 1% of the data, fell sharply to 2.60 points by 10%, and then stabilized around 2.6–2.9 points from 10% through 100%. This pattern is consistent with the stated hypothesis — the pretrained Transformer’s advantage is largest precisely in the low-data regime, where a classical model has too few examples to learn good features from scratch, while DistilBERT can lean on language structure learned during pretraining. Once enough labeled data is available (roughly 10% $\approx$ 12,000 examples and beyond), the marginal value of pretraining shrinks and the gap plateaus rather than continuing to widen.

4.3 Per-Class Performance (Full-Data DistilBERT Model)

Class	Precision	Recall	F1-score	Support
World	0.96	0.95	0.96	1,900

Class	Precision	Recall	F1-score	Support
Sports	0.99	0.99	0.99	1,900
Business	0.93	0.91	0.92	1,900
Sci/Tech	0.91	0.94	0.92	1,900

Performance is fairly balanced across classes (support is equal at 1,900 examples per class, so accuracy and macro-F1 are not distorted by class imbalance here). Sports is the easiest class ($F1 = 0.99$), while Business and Sci/Tech are the hardest ($F1 = 0.92$ each) and are the most frequent source of confusion with one another — unsurprising, since technology-company earnings and product news can plausibly be filed under either label.

5. Discussion & Observation

1. A 1-D convolution over word embeddings and a self-attention layer both build sentence representations. What can convolution “see” that attention cannot, and vice versa? Give a sentence on which you would expect the two to disagree, and say why.

A convolution applies a fixed-width sliding filter, so it directly captures local, position-invariant n-gram patterns (e.g. “not good” as a fused unit) regardless of where that n-gram sits in the sentence, but it cannot relate two words that fall outside its kernel width without stacking many layers. Self-attention has no built-in notion of adjacency: every token can attend to every other token in a single layer, so it captures long-range dependencies immediately (subject-verb agreement across a long clause, or a pronoun referring back to a distant noun) but has no inherent bias toward local structure — it has to learn locality, if it needs it, from data. On a sentence like “The company that missed its earnings target for three consecutive quarters finally turned a profit,” attention should easily connect “turned a profit” back to “The company” despite the long intervening clause, while a shallow convolution, limited by its receptive field, would likely need several stacked layers to bridge that same distance and may instead latch onto the strong local pattern “missed its earnings target” as the dominant (and here, misleading) signal for sentiment or topic.

2. Could you place a convolutional layer inside a Transformer block (or attention inside a text-CNN)? Where would you insert it, and what would each mechanism contribute to the final representation?

Yes — a natural hybrid inserts a depthwise convolution immediately before or in parallel with the self-attention sub-layer inside a Transformer block, so the model first pools local n-gram context and then lets attention route information globally over those locally-enriched tokens; this is the same intuition behind convolutional subword tokenizers used in some efficient Transformer variants. Symmetrically, one could add a lightweight self-attention layer on top of a text-CNN’s pooled feature maps, after several convolutional layers have already built up local n-gram detectors, so that attention aggregates across those detected patterns rather than raw tokens. In both placements, convolution contributes cheap, strongly-biased local composition, while attention contributes the ability to combine those local features across arbitrary distances in the sentence.

6. Limitations & Next Steps

- AG News is a relatively clean, well-separated dataset on which even a bag-of-words baseline performs strongly (83–92% accuracy). The data-efficiency gap observed here may understate the advantage of pretraining on noisier or more ambiguous text, such as GoEmotions or Jigsaw Toxic Comments.
- The TF-IDF baseline was refit independently at every data fraction rather than tuned per-fraction (e.g. re-selecting max_features), so its reported numbers are a lower bound on what a fully-tuned classical baseline could achieve at each size.
- A natural next step is repeating this sweep on a noisier, imbalanced dataset (GoEmotions) to test whether the low-data advantage of DistilBERT widens, as the limitations above would predict.